

Vector erase() in C++ STL

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In C++, vector **erase()** is a built-in function that is used to delete elements from the vector. It removes an element of a specific position or range of elements from the vector.

Let's take a simple example that uses the vector erase() method:

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<int> v = {1, 2, 3, 4, 5};

    // Remove the element at index 2
    v.erase(v.begin() + 2);

    for (auto i : v)
        cout << i << " ";
    return 0;
}
```

Output

```
1 2 4 5
```

In the above example, erase() method remove the element(**3**) which is present at index 2.

Syntax of Vector erase()

Vector erase method has two implementations:

```
// Remove single element
v.erase(pos);

// Erase range of elements
v.erase(first, last);
```

where,

Parameters:

- **pos:** Iterator to the element to be deleted.
- **first:** Iterator to the first element in the range.
- **last:** Iterator to the theoretical element just after the last element in the range.

Return Value:

- Returns an iterator pointing to the next element after deleted elements.
- If we delete the elements up to last element from the vector, then it will return an iterator to the [vector end\(\)](#).

Examples of Vector erase()

The following programs illustrates how to use the vector erase() method to remove elements in different cases:

Remove an Element Using Index from a Vector

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<int> v = {1, 2, 3, 4, 5};

    // Remove the element at index 1
    v.erase(v.begin() + 1);

    for (auto i : v)
        cout << i << " ";
    return 0;
}
```

Output

1 3 4 5

Explanation: The iterator of the element at index 1 is determined by adding the index to vector [begin\(\)](#) iterator. Then vector erase() is used to remove it.

Remove the Last Element from the Vector

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<int> v = {1, 2, 3, 4, 5};

    // Remove the last element
    v.erase(v.end() - 1);

    for (auto i : v)
        cout << i << " ";
    return 0;
}
```

Output

```
1 2 3 4
```

In the above program, we use [`end\(\)`](#) iterator to delete the last element of the vector, which is 5.

Remove a Range of Elements from a Vector

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    vector<int> v = {1, 2, 3, 4, 5};

    // Remove elements in the range [1, 4)
    v.erase(v.begin() + 1, v.begin() + 4);

    for (auto i : v)
        cout << i << " ";
    return 0;
}
```

Output

```
1 5
```

Explanation: The iterator to the part of the vector to be removed is passed to the `vector erase()` method. It removed that part from the array.

Vector `clear()` and `erase()`

In a vector, both **`vector clear()`** and **`vector erase()`** are used for element removal, but they serve different purposes. Following are the cases which describe when to use `clear()` and when to use `erase()`:

- If we have to remove all elements from the vector, then we use **`vector clear()`** method.
- If we have to remove some specific elements from the vector, then we use **`vector erase()`** method.

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